

MAKINDU, Kenya

WMO: 637660

Lat: 2.2841S

Lon: 37.8208E

Elev: 1000

StdP: 89.87

Time Zone: 3.00 (E03)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	14.6	8.7	7.9	25.2	9.7	8.4	25.4	8.2	24.4	6.9	23.2	1.3	270	0.259

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	11.3	32.3	19.4	31.6	19.5	30.9	19.6	22.1	27.3	21.7	26.8	21.4	26.4	3.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	17.5	23.1	20.3	17.0	22.6	20.2	16.8	22.5	69.9	27.4	68.6	26.9	67.4	26.6	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	5.7	4.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	23.2	23.6	24.9	25.2	24.5	23.1	21.7	21.0	21.3	22.7	23.9	23.3	23.4	(6)
(7)		DBStd	1.77	1.23	1.17	1.17	1.16	0.94	1.09	1.10	1.36	1.30	1.18	1.40	1.22	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	2	0	0	0	0	0	0	0	2	0	0	0	0	(9)
(10)		CDD10.0	4819	421	418	471	434	406	351	341	350	382	430	399	415	(10)
(11)		CDD18.3	1778	163	184	213	184	148	101	83	93	132	172	149	157	(11)
(12)		CDH23.3	12037	1133	1586	1777	1238	913	609	435	592	866	1330	804	754	(12)
(13)		CDH26.7	3278	328	631	683	346	182	79	35	89	197	420	171	118	(13)
(14)	Wind	WSAvg	2.2	2.1	2.0	2.2	2.2	1.9	2.1	2.2	2.4	2.8	2.8	2.1	1.8	(14)
(15)	Precipitation	PrecAvg	581	38	29	62	110	24	2	1	2	31	180	109		(15)
(16)		PrecMax	1208	190	135	205	287	98	20	6	18	14	222	478	318	(16)
(17)		PrecMin	256	0	0	0	1	0	0	0	0	0	0	16	3	(17)
(18)		PrecStd	247	43	32	58	77	23	4	1	3	4	47	117	76	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.8	33.5	33.1	32.1	30.1	29.4	29.0	29.9	30.8	31.9	31.1	30.0	(19)
MCWB			18.7	19.5	19.6	20.0	20.2	18.9	17.3	17.6	18.0	19.3	19.8	20.6	(20)	
2%		DB	31.2	32.5	32.2	30.7	29.2	28.2	27.9	28.6	29.8	30.9	29.9	28.9	(21)	
		MCWB	19.1	19.3	19.6	20.3	19.9	18.4	17.5	17.5	17.8	19.0	19.7	20.8	(22)	
5%		DB	30.0	31.6	31.3	29.8	28.6	27.5	26.9	27.6	28.9	30.0	28.8	28.0	(23)	
		MCWB	19.1	19.4	19.8	20.3	19.8	18.3	17.3	17.4	17.8	18.7	19.8	20.8	(24)	
10%		DB	28.7	30.5	30.4	28.9	27.9	26.8	25.9	26.7	27.9	29.0	27.6	27.1	(25)	
		MCWB	19.6	19.6	19.9	20.3	19.7	18.1	17.0	17.3	17.7	18.5	20.0	20.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.1	22.2	22.4	22.4	22.1	21.1	19.5	20.3	20.1	21.2	21.9	22.2	(27)
MCD B			26.8	28.7	28.1	27.4	27.2	26.6	26.0	26.3	26.8	27.0	26.7	26.9	(28)	
2%		WB	21.5	21.6	21.7	21.7	21.5	19.8	18.4	18.9	19.2	20.4	21.4	21.7	(29)	
		MCD B	26.1	27.9	27.3	26.8	26.6	25.3	24.7	25.0	26.0	25.9	25.9	26.3	(30)	
5%		WB	21.0	21.1	21.4	21.3	21.0	19.2	18.0	18.2	18.7	19.9	21.0	21.4	(31)	
		MCD B	25.7	27.3	26.7	26.4	26.0	24.5	24.0	24.6	25.3	25.3	25.3	25.8	(32)	
10%		WB	20.5	20.7	21.0	21.0	20.6	18.8	17.6	17.7	18.3	19.6	20.7	21.1	(33)	
		MCD B	25.3	26.6	26.2	26.0	25.3	23.9	23.5	24.1	24.7	25.1	24.7	25.2	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.8	12.0	11.3	9.9	10.0	10.7	10.8	11.5	11.3	11.5	8.7	8.1	(35)
MCDBR			13.3	13.7	12.6	10.9	11.3	12.3	12.2	13.1	12.5	12.6	10.8	9.3	(36)	
MCWBR			3.9	3.4	3.1	2.6	3.3	3.9	3.5	3.8	3.0	2.9	2.6	2.6	(37)	
5% WB		MCDBR	9.7	11.5	10.8	9.8	9.5	10.5	10.6	11.9	11.3	11.1	8.4	8.3	(38)	
		MCWBR	3.8	3.3	3.2	2.9	3.5	3.7	3.5	4.2	3.2	3.1	2.8	2.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.399	0.419	0.398	0.349	0.343	0.376	0.388	0.373	0.360	0.364	0.371	0.393	(40)	
taud		2.402	2.327	2.416	2.599	2.612	2.463	2.392	2.424	2.461	2.472	2.493	2.441	(41)		
Ebn at Noon		934	915	926	949	931	884	879	913	947	956	954	935	(42)		
Edh at Noon		124	136	123	99	94	107	116	116	116	116	113	119	(43)		
(44)	All-Sky Solar Radiation	RadAvg	6.26	6.53	6.37	5.72	5.12	4.60	4.57	4.89	5.82	6.11	5.79	5.91	(44)	
(45)		RadStd	0.35	0.30	0.38	0.26	0.19	0.27	0.38	0.35	0.34	0.30	0.24	0.29	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.47	+0.69	N/A	+0.57	+0.39	+0.16	N/A	N/A	N/A	N/A

Nomenclature: See separate page